Title: A Comparative Study of Bosonic Cat-qubit Codes for Quantum Chemistry Simulation

Year: 2024

Authors: Y. Petrov, P. Okafor (School of Information Sciences, Lakeshore University)

Keywords: NISQ, error correction, Hamiltonian simulation, quantum annealing, quantum gates

Abstract:

Scalable qubit connectivity remains a critical challenge for Hamiltonian simulation, particularly in quantum chemistry simulation. This paper proposes a framework built on bosonic cat-qubit codes that explicitly accounts for crosstalk reduction in multi-qubit devices. We evaluate the framework on the MQT Bench benchmark collection and compare it against 4 baseline methods. Experimental results show that the proposed approach improves logical error rate by 32.1% while keeping approximation ratio within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of Hamiltonian simulation has changed how researchers and practitioners approach scalable qubit connectivity, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Over the past decade, bosonic cat-qubit codes have attracted sustained attention as a promising route toward solving scalable qubit connectivity in realistic settings such as quantum chemistry simulation. Despite considerable progress, existing solutions for scalable qubit connectivity tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. This paper addresses these gaps by studying how bosonic cat-qubit codes interact with neutral-atom qubit arrays when applied to scalable qubit connectivity.

Our work is motivated by the growing demand for quantum chemistry simulation, where reliability and transparency requirements are especially strict. We make three contributions: a unified formulation of scalable qubit connectivity, a practical design built on bosonic cat-qubit codes, and an empirical evaluation on the MQT Bench benchmark collection. Understanding crosstalk reduction in multi-qubit devices is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

Subsequent work by Fernandez et al. (2013) extended this line of research to quantum chemistry simulation, although the evaluation was restricted to small-scale settings. A survey by Novak et al. (2015) categorised existing approaches to scalable qubit connectivity into three families and highlighted the lack of standard benchmarks. Researchers working on Hamiltonian simulation have explored neutral-atom qubit arrays as an alternative, with Banerjee et al. (2023) noting that the approach scales poorly beyond 1200 components. OConnor et al. (2020) analysed crosstalk reduction in multi-qubit devices and showed that purely heuristic solutions cannot guarantee consistent approximation ratio. More recently, Larsen et al. (2014) combined bosonic cat-qubit codes with neutral-atom qubit arrays, achieving a 38.7% reduction in overhead on the MQT Bench benchmark collection.

Several studies have criticised the reliance on synthetic workloads, arguing that results on the MQT Bench benchmark collection may not generalise to quantum chemistry simulation. In contrast to these efforts, our work treats scalable qubit connectivity and crosstalk reduction in multi-qubit devices jointly rather than as independent concerns. Industry reports corroborate these academic findings, pointing to scalable qubit

connectivity as a persistent obstacle for quantum chemistry simulation. Despite this body of literature, no prior study has systematically compared bosonic cat-qubit codes and neutral-atom qubit arrays under identical experimental conditions.

3. Methodology

We formalise scalable qubit connectivity as a constrained optimisation task in which the objective is to maximise logical error rate while keeping gate fidelity within a prescribed budget. The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on bosonic cat-qubit codes, and a validation layer that incorporates neutral-atom qubit arrays. All experiments were conducted on the MQT Bench benchmark collection, which provides 1200 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 10 times with independent random seeds, and we report mean values together with 95% confidence intervals.

Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data. The core routine applies bosonic cat-qubit codes to reduce the effect of crosstalk reduction in multi-qubit devices, followed by a calibration step that stabilises logical error rate. Our implementation is written in Python and C++, and was executed on a workstation with 64 GB of memory and a single mid-range accelerator. We compare against 4 baseline methods, including a conventional approach built on neutral-atom qubit arrays and two recent variants reported in the literature.

Ablation experiments show that removing the neutral-atom qubit arrays component reduces logical error rate by approximately 8.0%, confirming its contribution. A sensitivity analysis over 32 parameter settings shows that performance remains stable, with variation in approximation ratio below 3.3%. Threats to validity include the limited diversity of the MQT Bench benchmark collection and the possibility that deployment in quantum chemistry simulation introduces conditions not captured in our setup.

4. Conclusion

This paper investigated scalable qubit connectivity through the lens of bosonic cat-qubit codes, and demonstrated that a carefully designed pipeline can improve logical error rate by 32.1% on the MQT Bench benchmark collection. The findings suggest that Hamiltonian simulation will continue to benefit from closer integration of bosonic cat-qubit codes and neutral-atom qubit arrays, particularly in quantum chemistry simulation. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of crosstalk reduction in multi-qubit devices. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for scalable qubit connectivity.